MLA01 - Artificial Intelligence and Expert System Laboratory - List of Experiments			
Exp. No.	**Scenario**	**Laboratory Experiment**	**Expected Outcome**
1	A robot must find the shortest route to visit all rooms in a building level by level.	Implement Breadth First Search (BFS) Algorithm.	Explore nodes level-wise and find the shortest path in an unweighted graph.
2	A rescue drone explores a cave by moving as deep as possible before backtracking.	Implement Depth First Search (DFS) Algorithm.	Traverse nodes using depth-first exploration.
3	A logistics company wants to deliver goods with the minimum transportation cost.	Implement Uniform Cost Search (UCS) Algorithm.	Find the least-cost path between source and destination.
4	A GPS navigation system selects the next city based on the estimated distance to the destination.	Implement Best First Search Algorithm.	Find a path using heuristic evaluation.
5	A warehouse robot chooses the nearest storage rack at every step to minimize travel time.	Implement Greedy Best First Search Algorithm.	Solve pathfinding using greedy heuristics.
6	A delivery robot must reach its destination using both actual travel cost and estimated remaining distance.	Implement A* Search Algorithm.	Obtain the optimal path using heuristic and path cost.
7	A chess-playing AI must determine the best move against an opponent.	Implement the Minimax Algorithm.	Select the optimal move through adversarial search.
8	A game AI needs to make faster decisions by avoiding unnecessary game-tree exploration.	Implement Alpha-Beta Pruning.	Reduce search time while preserving optimal decisions.
9	A person has two containers of different capacities and must measure an exact quantity of water.	Solve the Water Jug Problem.	Generate the sequence of operations to reach the goal state.
10	A robotic arm transfers stacked containers while following movement constraints.	Solve the Towers of Hanoi Problem.	Generate the optimal sequence of moves.
11	A banking application verifies encrypted arithmetic puzzles before processing transactions.	Solve a Crypt Arithmetic Problem.	Assign valid digits satisfying arithmetic constraints.
12	A bank wants to approve or reject loan applications based on customer information.	Build a Decision Tree Classifier.	Classify data using decision tree learning.
13	A hospital predicts whether a patient has a disease using historical medical records.	Implement an Artificial Neural Network.	Train and test a neural network model.
14	A nutrition application recommends diet plans based on age, BMI, and health conditions.	Develop a Diet Recommendation Expert System.	Recommend personalized diet plans using rules.
15	A genealogy application determines relationships among family members.	Implement Family Knowledge Representation.	Infer family relationships using logical rules.
16	A wildlife information system determines whether an animal can fly based on its characteristics.	Implement the Birds Fly Knowledge Base.	Perform inheritance-based reasoning.
17	A healthcare expert system assists doctors by identifying diseases from patient symptoms.	Develop a Medical Expert System.	Infer diseases using rule-based reasoning.
18	A fruit-sorting machine identifies fruits based on their colors.	Implement Fruit Color Classification.	Classify fruits using production rules.
19	An astronomy application answers questions about planets in the solar system.	Implement Planet Knowledge Representation.	Retrieve planetary facts using logical inference.
20	A citizen information portal retrieves personal details using date of birth records.	Implement Person DOB Knowledge Base.	Query stored facts using logical reasoning.
21	An image-processing system identifies recurring visual patterns automatically.	Implement Pattern Recognition.	Detect patterns using AI techniques.
22	A robot must retrieve food placed on a high shelf by planning a sequence of actions.	Solve the Monkey and Banana Problem.	Generate an optimal action plan.
23	A technical support system diagnoses computer faults by applying known rules to user inputs.	Implement Forward Chaining.	Derive conclusions from available facts.

24	A medical diagnosis system determines the possible disease by working backward from symptoms.	Implement Backward Chaining.	Reach conclusions through goal-driven reasoning.
25	A search application written in Prolog locates the best route using heuristic information.	Implement Best First Search in Prolog.	Perform heuristic search using Prolog.
26	A mathematical tutoring system computes the sum of natural numbers recursively.	Implement Sum of 1 to N using recursion.	Demonstrate recursive problem solving.
27	A text-processing application identifies vowels in a given sentence for language analysis.	Implement Vowel Identification using rule-based logic.	Classify characters based on predefined rules.